



Predicting Credit Card Churn: Identifying At-Risk Customers

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What if we could identify credit card customers who are about to leave before they do?

Every month, banks lose millions in revenue from customers who quietly cancel their credit cards.

What if you could predict this before it happens?



Credit Card churn is a hidden revenue loss



LOSING CREDIT CARD CUSTOMERS MEANS LOSING:

- Transaction revenue
- Interest income
- Long-term customer value

Credit cards are more than payment tools



DAILY SPENDING BEHAVIOR

- Frequent transactions create continuous revenue streams

CUSTOMER RELATIONSHIP

- Card usage reflects customer engagement with the bank

GROWTH OPPORTUNITIES

- Active customers create opportunities for:
 - Loans
 - Insurance products
 - Other banking services



The AI-Powered Solution



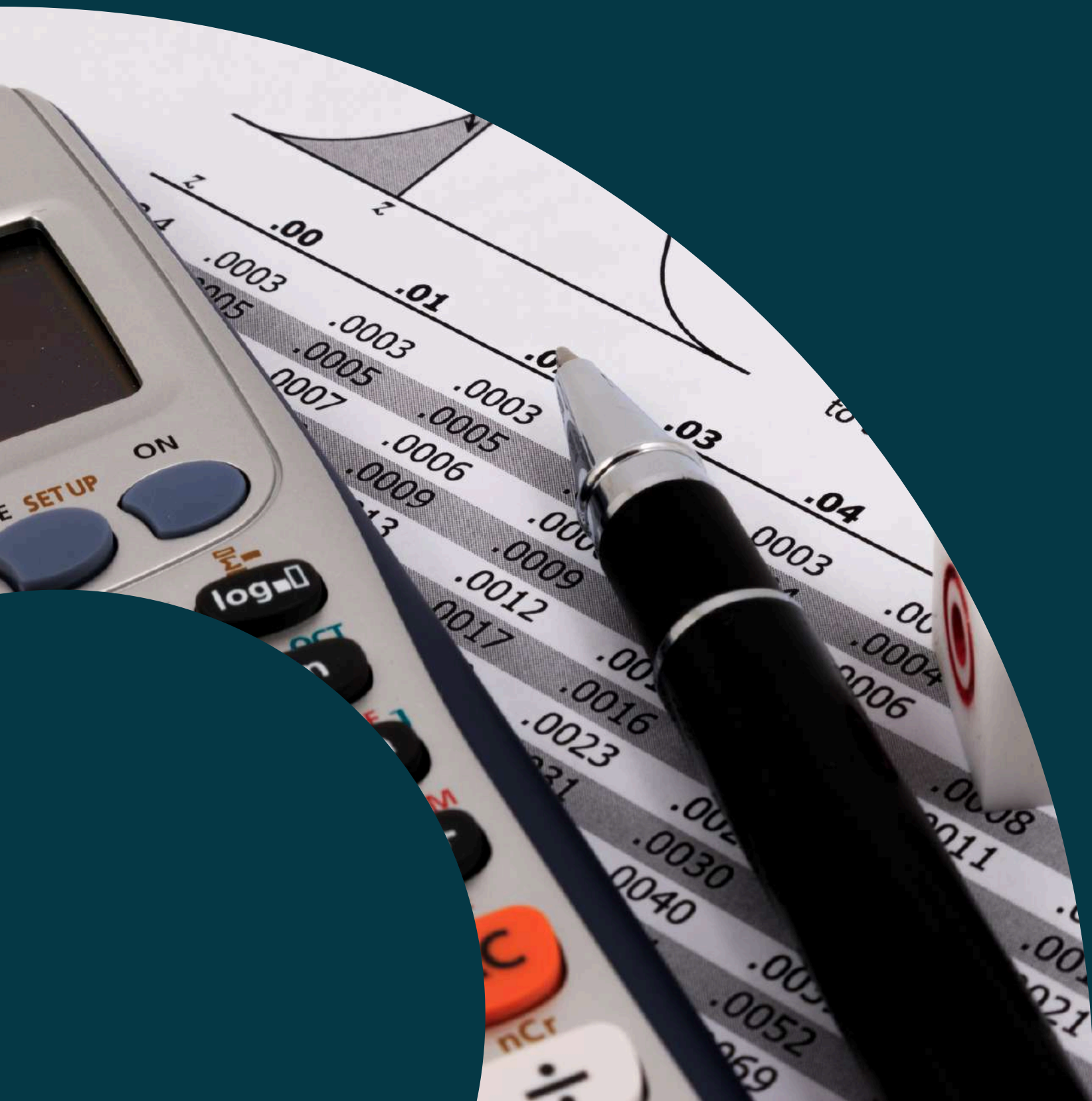
1. ANALYZE CUSTOMER BEHAVIOR

Machine learning analyzes customer data, spending patterns, and engagement activity to understand customer behavior.



2. PREDICT CHURN RISK

The model identifies customers who show patterns indicating a high probability of leaving.



3. PRIORITIZE HIGH-RISK CUSTOMERS

The bank focuses its attention on customers with the highest churn probability instead of targeting everyone.



PRIORITIES

A SNAPSHOT OF THE CUSTOMER BASE

20,000 customer accounts · 55 raw data fields · 12 months of transaction history



Demographics & Profile

Age, marital status, education, employment, household size, residence type.



Spending & Transactions

Monthly and yearly spend, transaction counts, average ticket size, category mix.



Credit & Risk Signals

Credit limit, utilization, late-payment history, spend volatility, card tenure.



Loyalty & Engagement

Loyalty score, active days, product holdings, account types, balances.



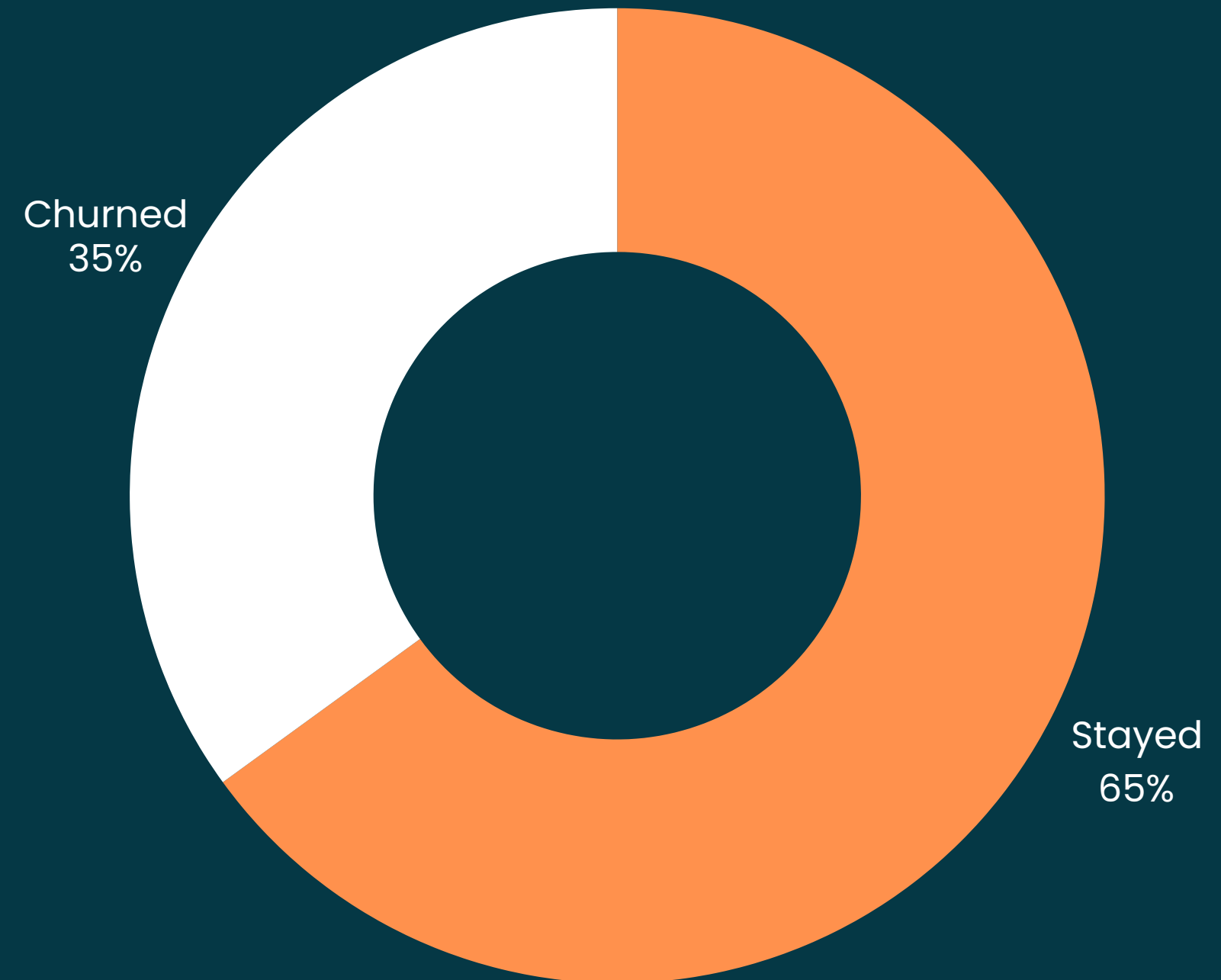
Understanding The Data



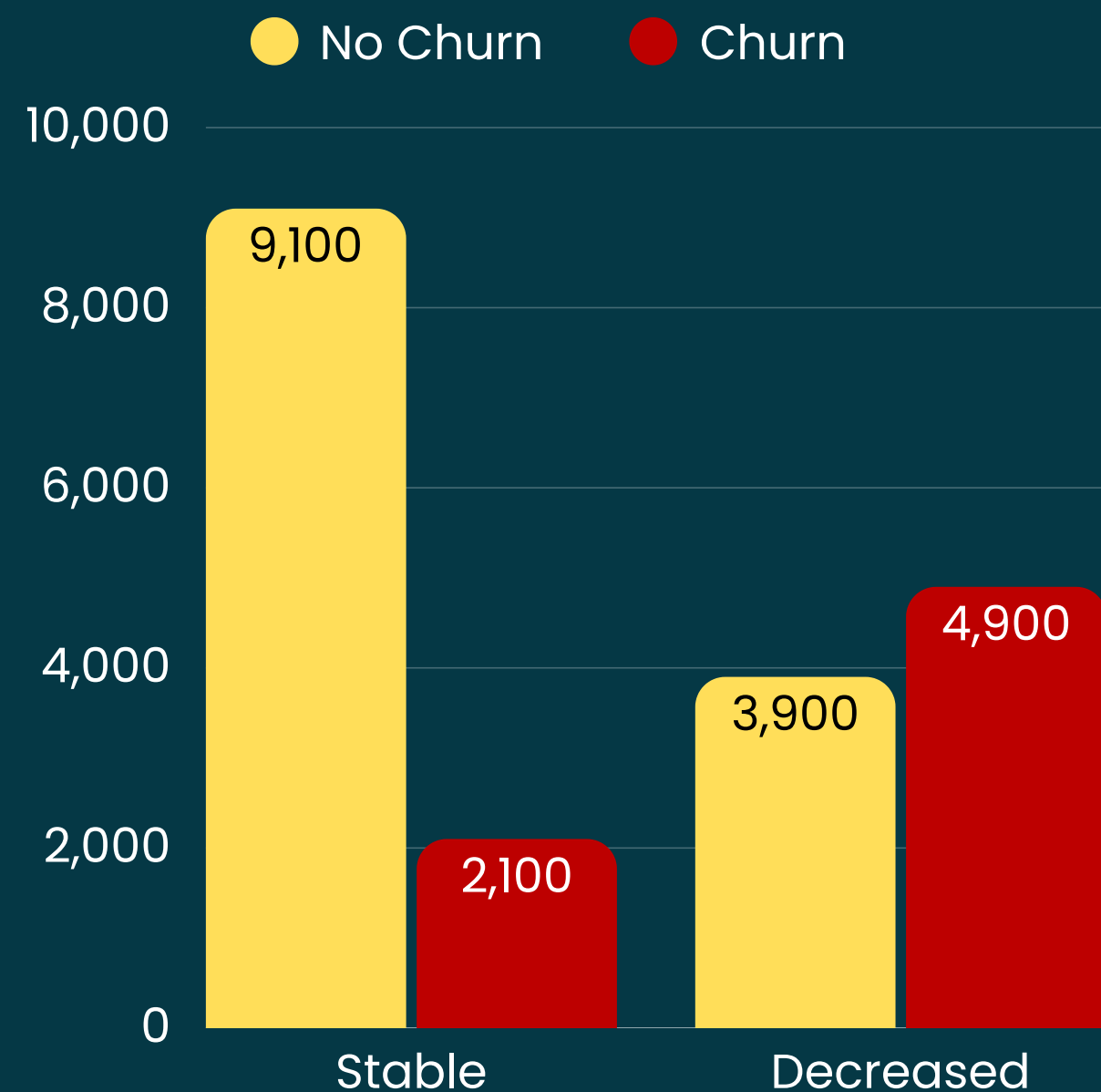
In this case
study we had

35%
Churn
Rate

FOR CREDIT CARD ACCOUNTS

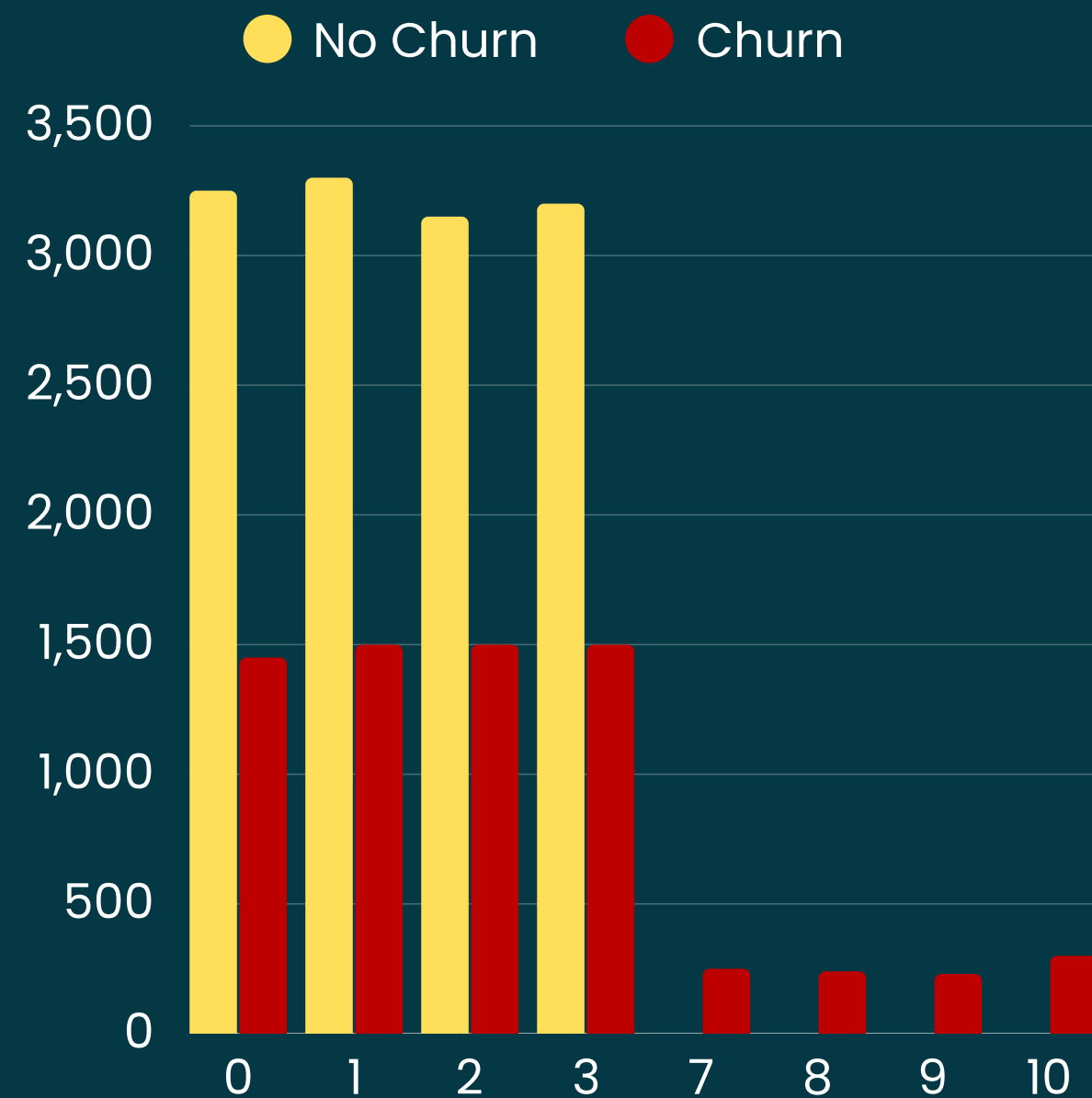


Decreasing Spending Trend

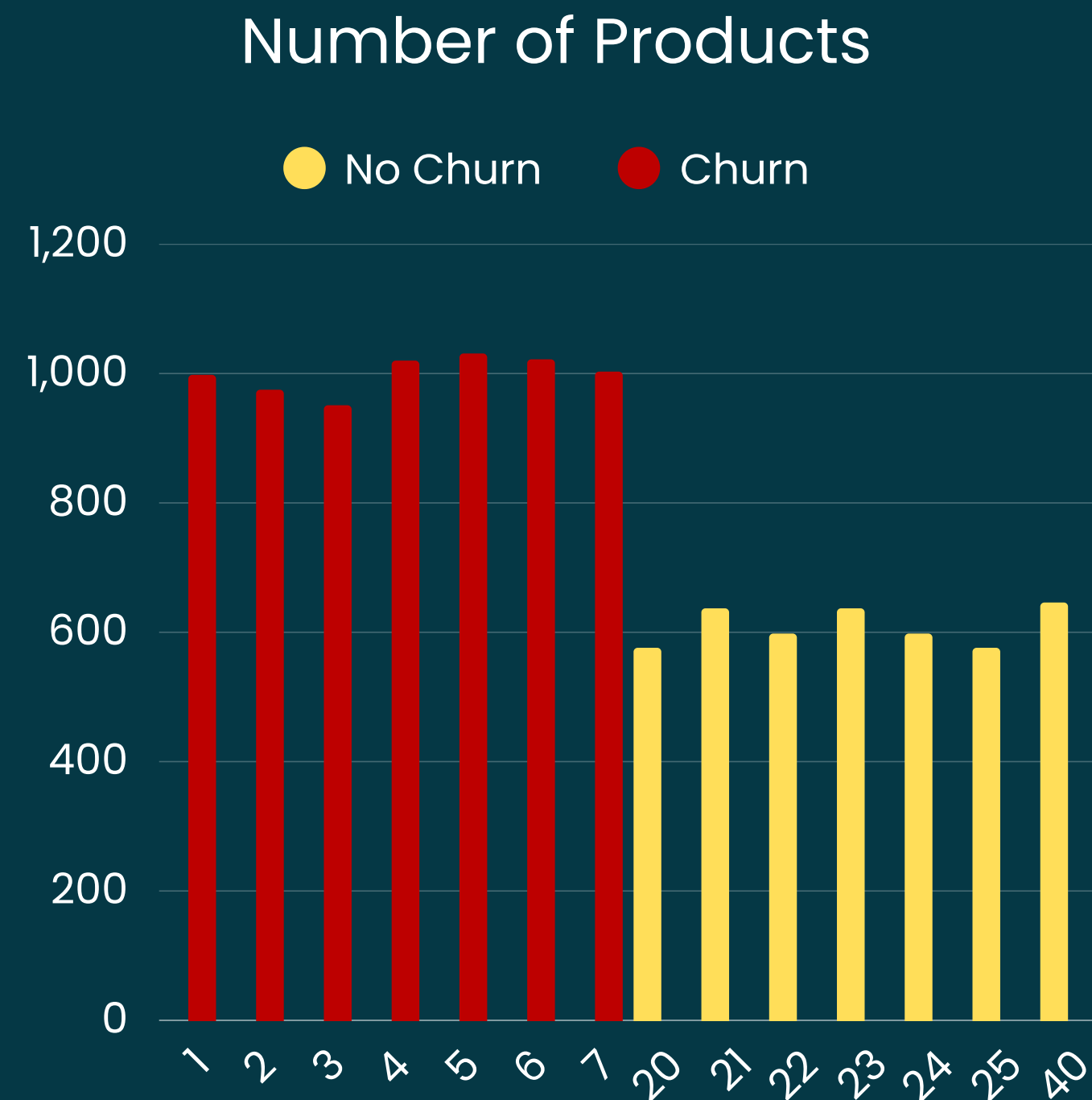


Customers with
a decreased
spending rate
Tends to leave.

Late Payments Frequency



Customers with 7 or more late payments have a 100% churn rate.



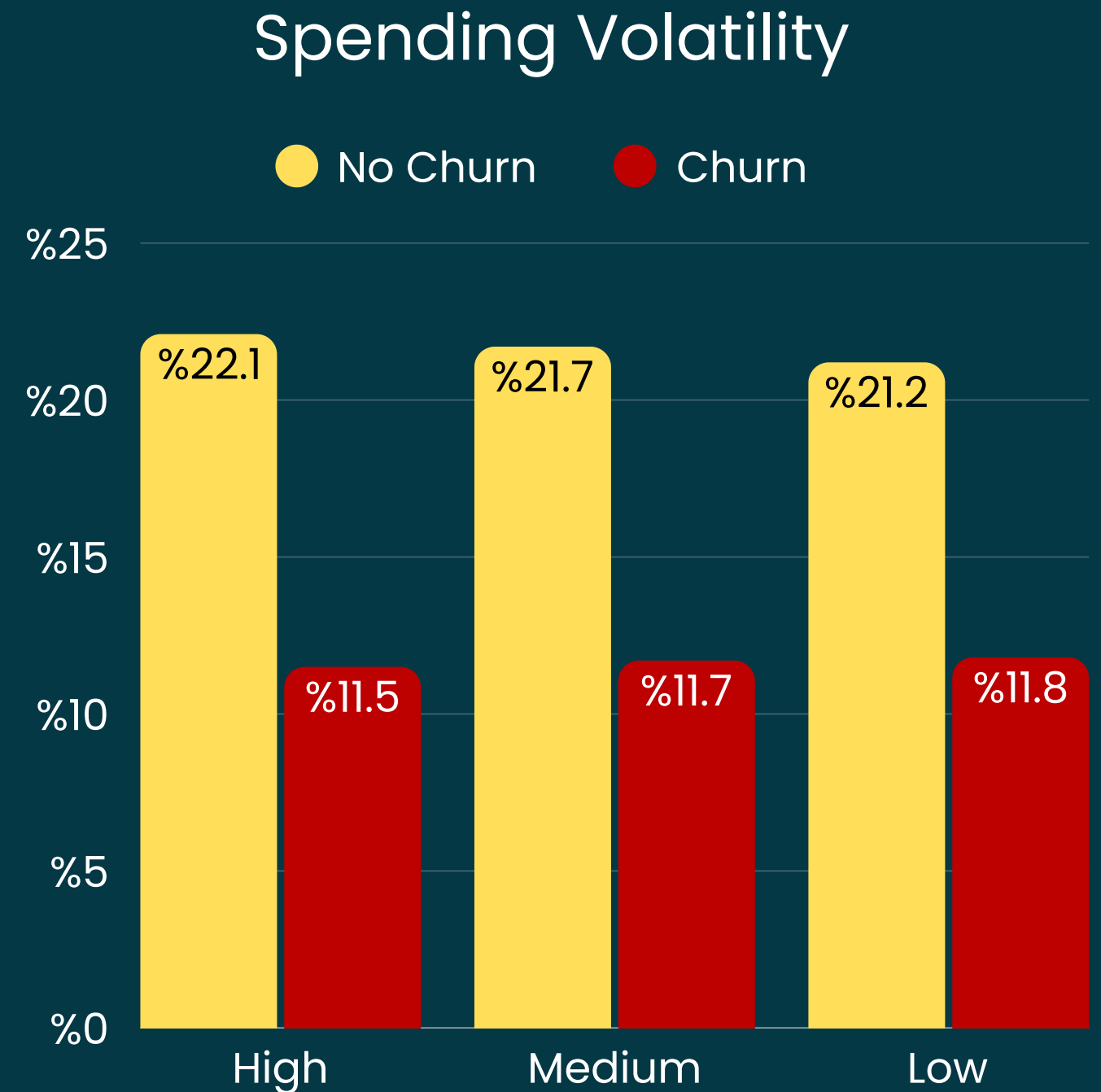
NUMBER OF PRODUCTS RELATION TO CHURN

Customers with 1–7 products uniformly churned, while those with 20–40 products entirely stayed.

This perfect, unnatural split is a textbook data leakage risk that must be excluded from the predictive model.

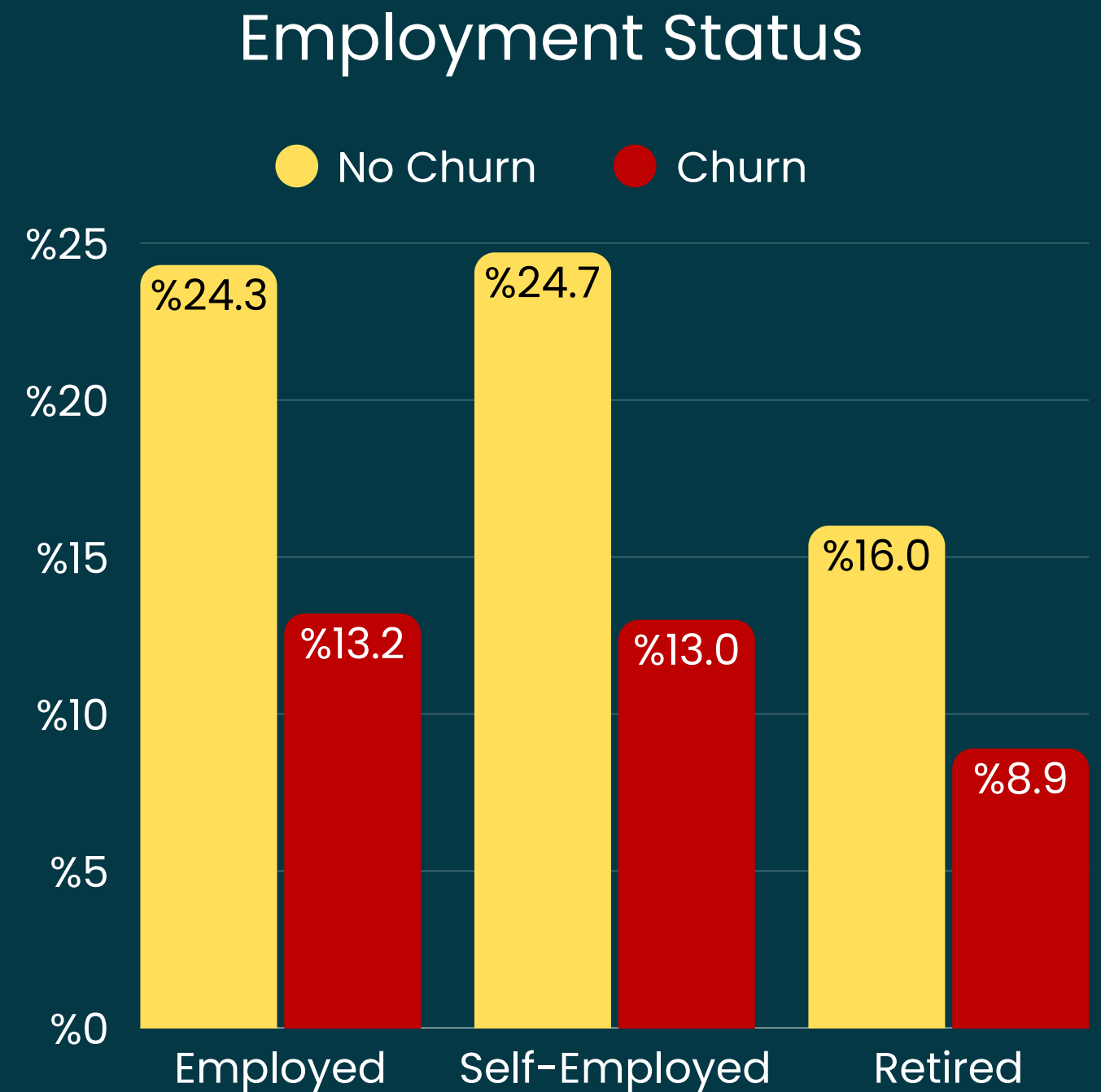
DOES IRREGULAR SPENDING SIGNAL CHURN RISK?

- Big changes in monthly bills **do not** make customers leave.
- Up-and-down spending is safe. We only need to worry when their total spending goes down and stays down.



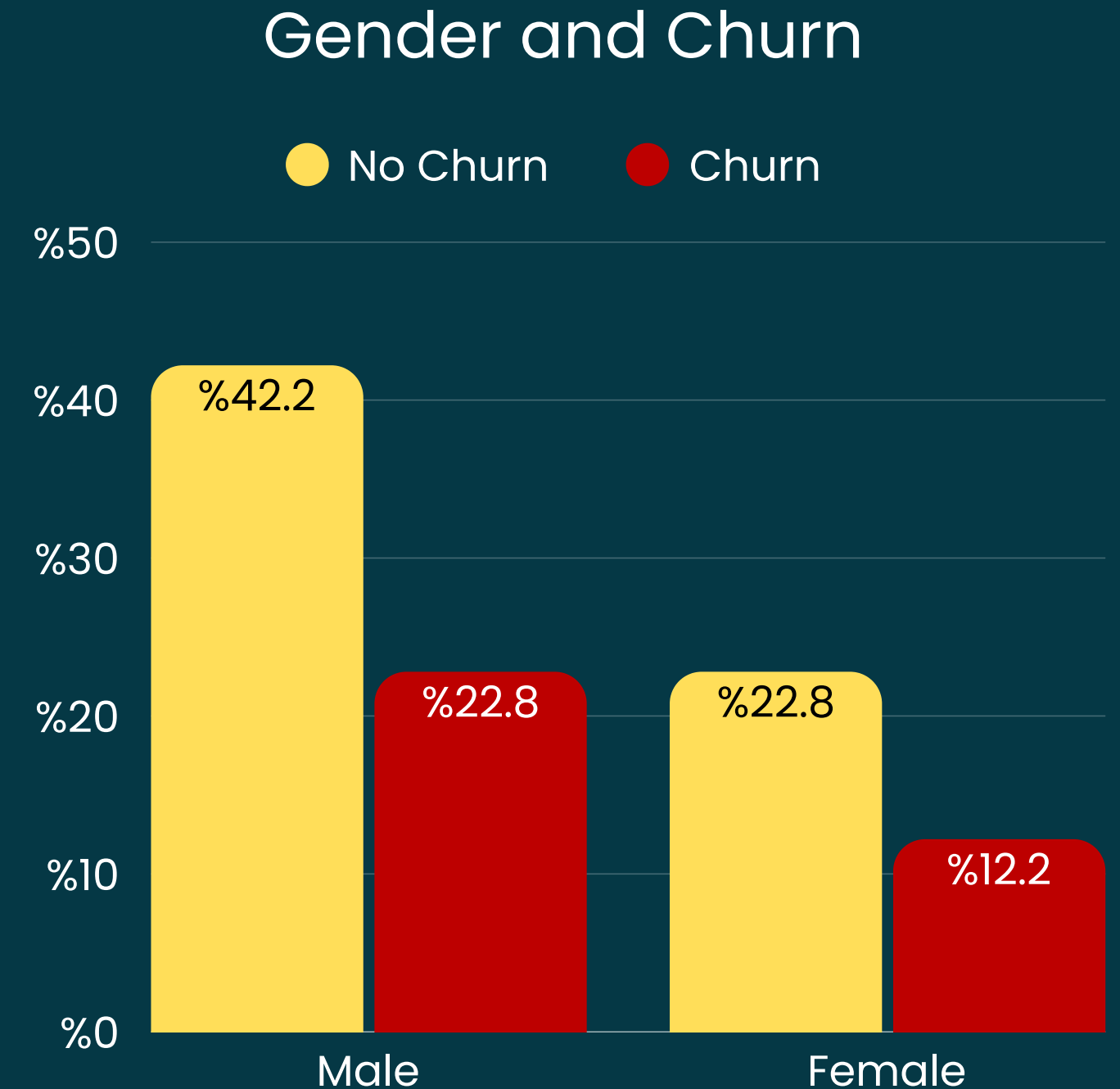
DOES JOB TYPE MATTER?

- A customer's job (employed, retired, or self-employed) does not change if they stay or leave.
- Everyone leaves at the exact same rate (~35%), so we shouldn't use job status to predict churn.



DOES THE GENDER AFFECT CHURN?

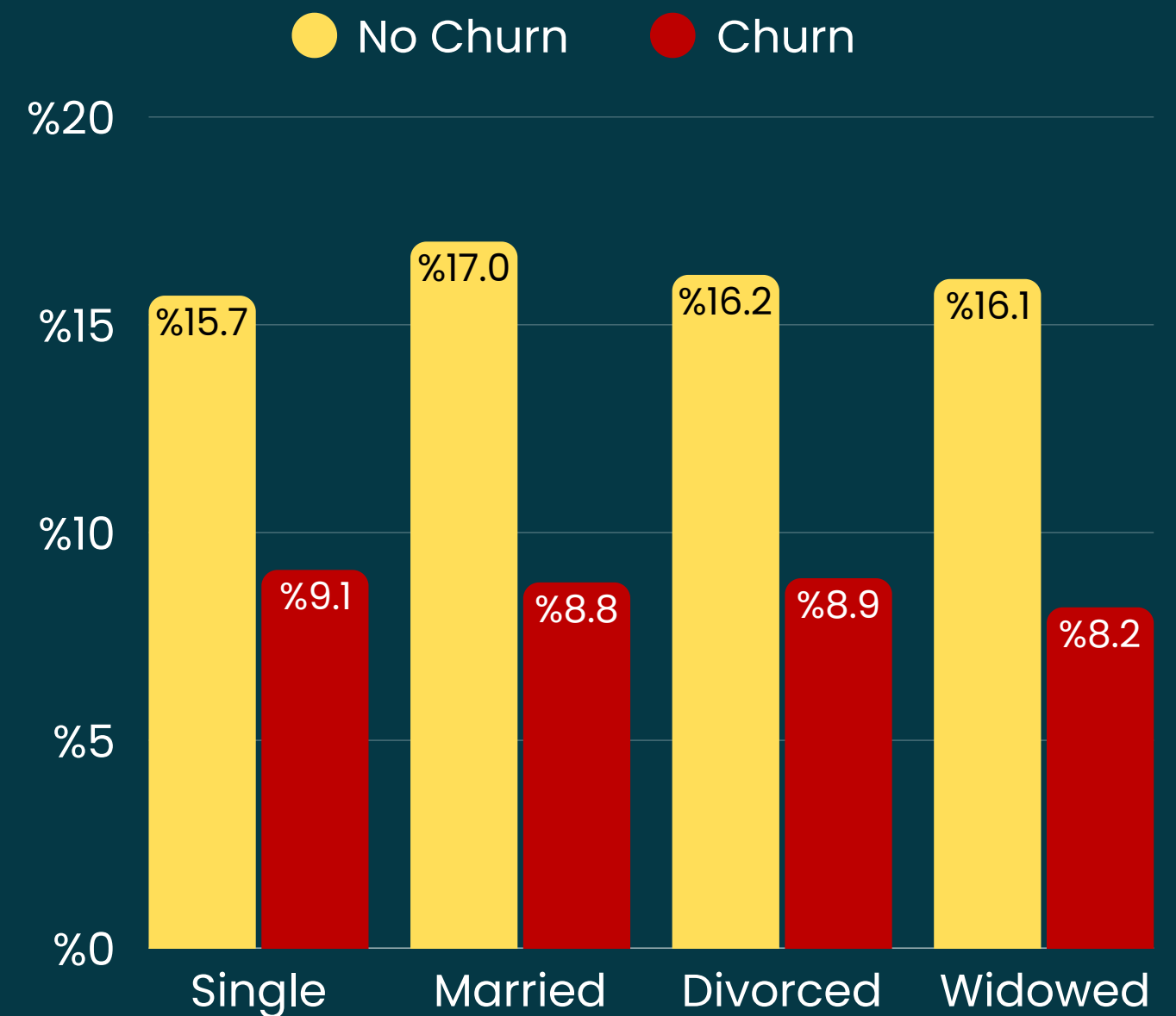
- A customer's gender (Male or Female) does not change if they stay or leave.
- Both men and women leave at the exact same rate (~35%).



MARITAL STATUS

- A customer's marital status (single, married, divorced, or widowed) does not change if they stay or leave.
- Everyone leaves at the exact same rate (~35%), so we shouldn't use marital status to predict churn.

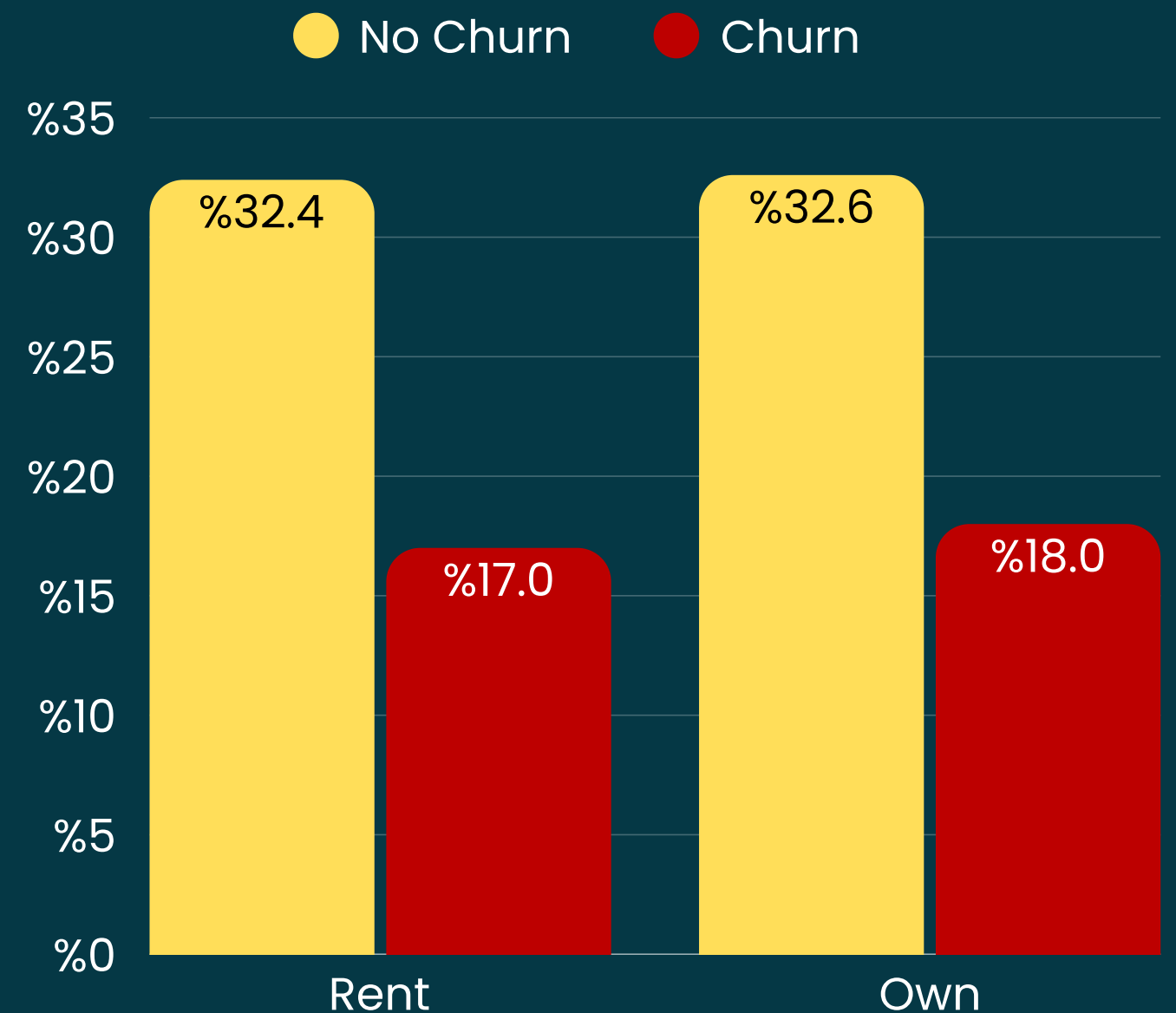
Marital Status and Churn



RESIDENCE TYPE

- A customer's residence type (rent or own) does not change if they stay or leave.
- Everyone leaves at the exact same rate

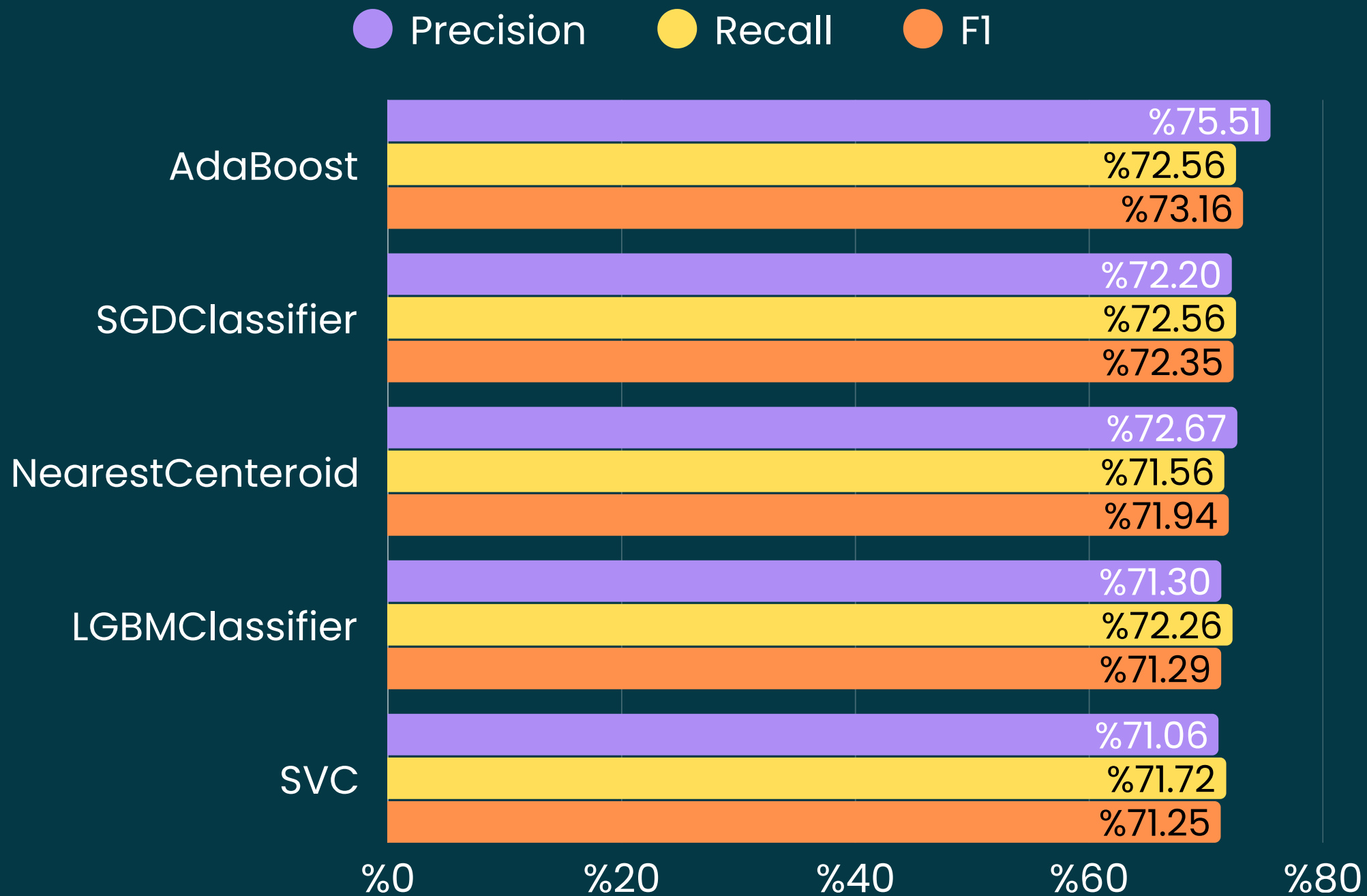
Residence Type and Churn





FROM RAW RECORDS TO A CLEAN, TRUSTWORTHY MODELING DATASET

- **Cleaned missing values:** Filled gaps in balances, spend and loyalty scores using median/mean. Kept missing education as "Unknown".
 - **Removed noise:** Dropped account numbers, masked card numbers and anonymized fields lacking predictive value.
 - **Engineered new signal:** Added "days to card expiration" as a new feature for renewal triggers.
 - **Encoded for the model:** Converted ordered and category fields (like gender, marital status and account type) into numeric formats.
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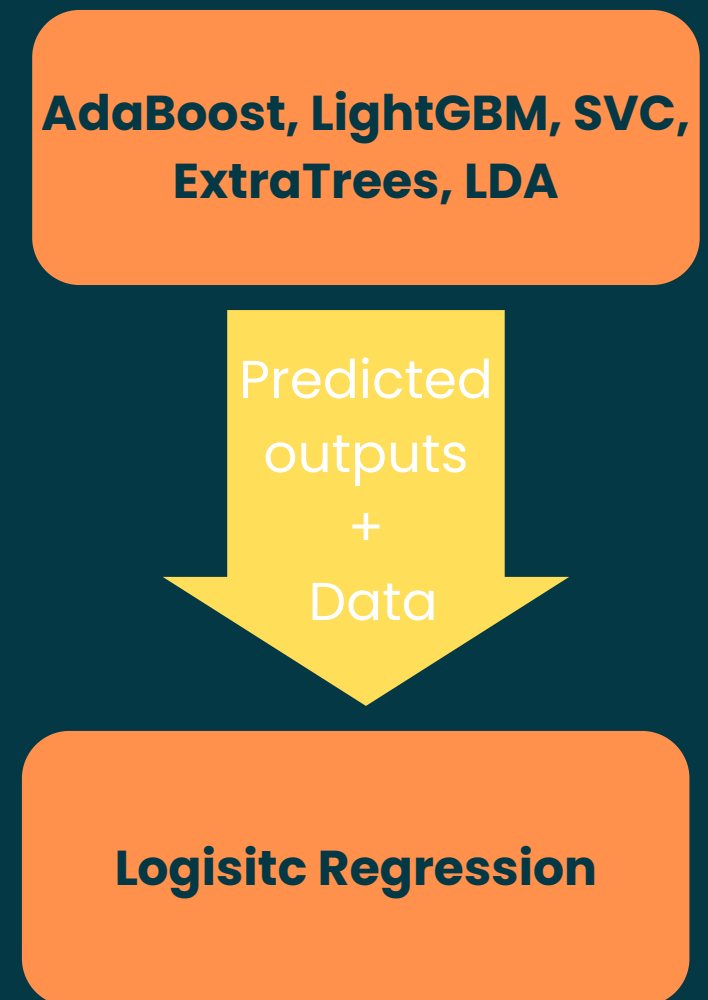
MODELLING TRIALS

25+ classifiers were benchmarked with LazyPredict before committing to an approach.

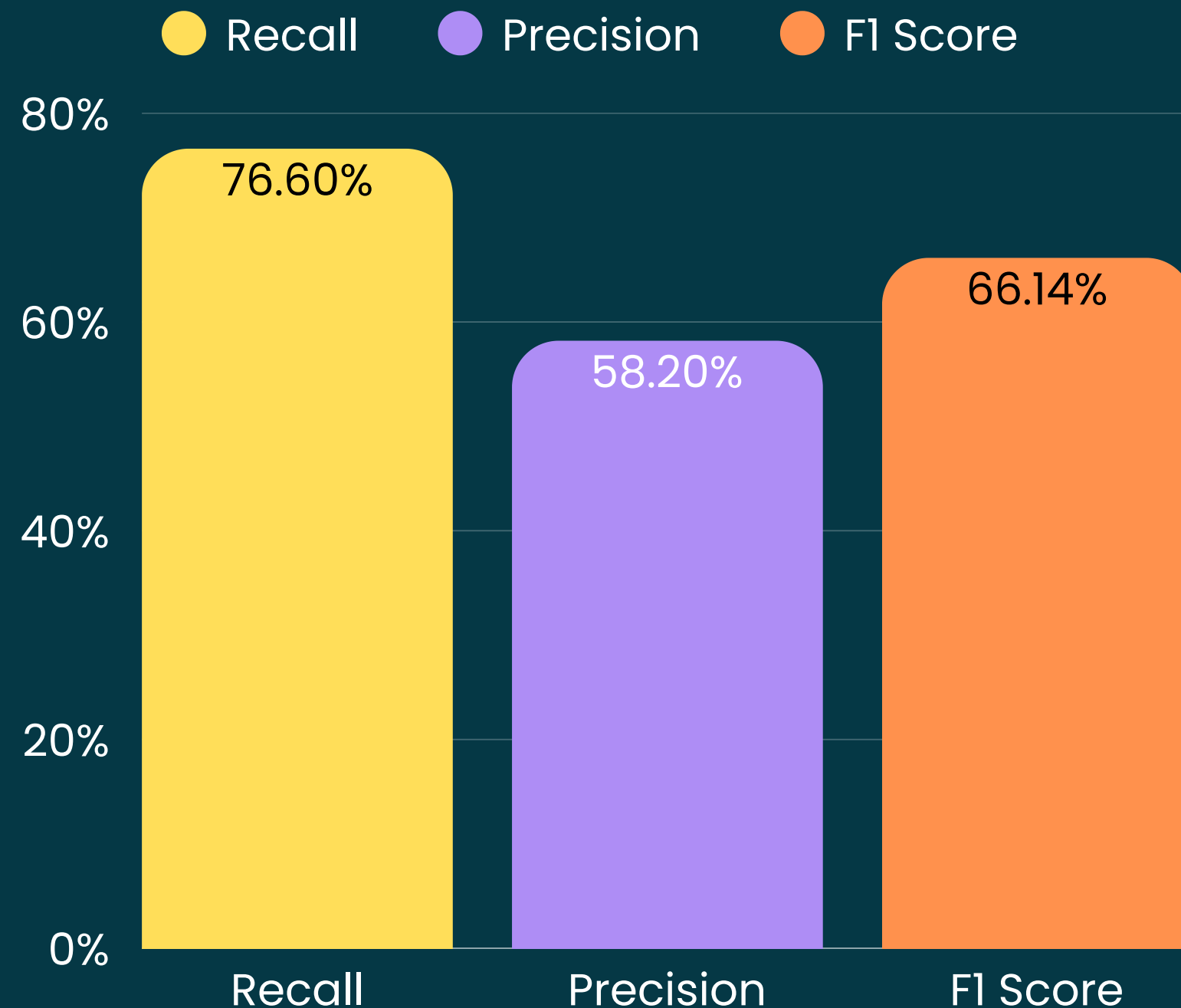


Why AdaBoost wasn't the finish line?

AdaBoost led on precision and F1, but a 5-model stacking ensemble was built on top of it to squeeze out more stable recall on the minority churn class.



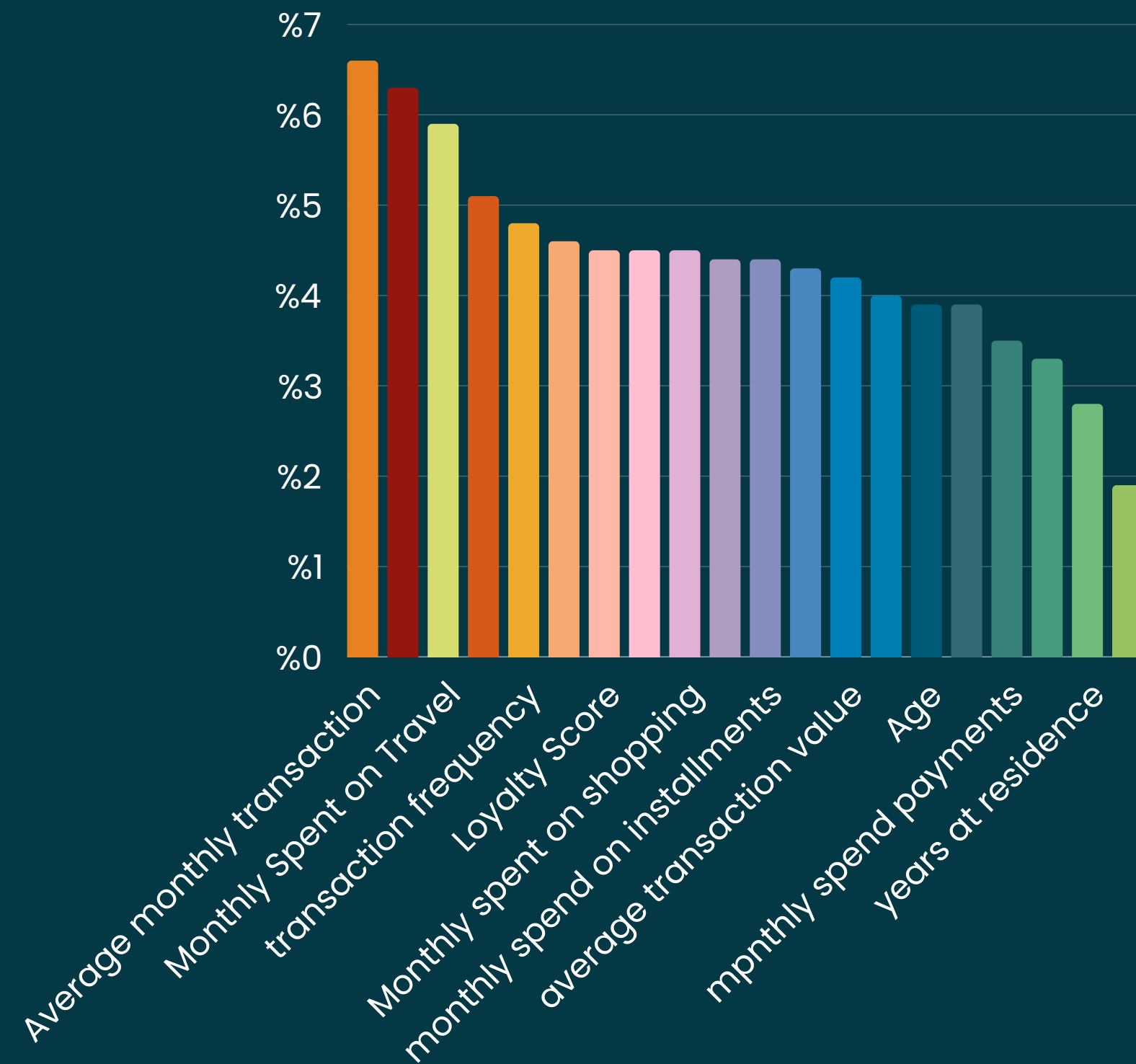
After Stacking



Model prioritizes recall to identify potential churners and support proactive retention strategies.

- Recall (76.60%): Identified 77% of actual churn customers.
- Precision (58.20%): 58% of predicted churners were actual churn risks.
- F1 Score (66.14%): Shows the balance between finding churn customers and avoiding unnecessary predictions.

Top-20 most Important Factors



**MOST
SIGNIFICANT
FACTORS FOR
CHURNING**



Future Work

- **Expand Data:** Train on a much larger dataset to enhance quality.
 - **Alternative AI Methods:** Use unsupervised learning like clustering to discover hidden customer churn patterns.
 - **Automated Alerts:** Notify teams when customer activity shows signs of churn.
 - **Targeted Offers:** Provide personalized retention strategies for high-risk customers.
 - **Enhanced Feature Engineering:** Develop deeper customer behavior insights.
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THANK YOU